

GCOp-amp

Fully-Differential Class-AB Folded-Cascode Operational Amplifier

Process: SkyWater SKY130A · **Supply:** 1.8 V · **Generated:** 2026-07-28

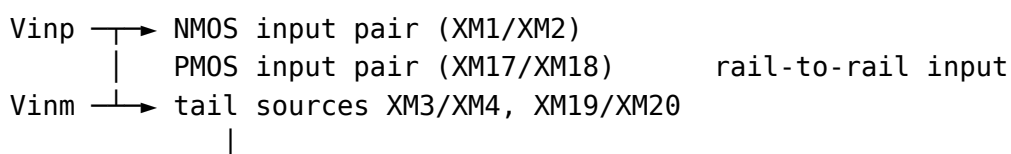
1 Features

- Fully-differential architecture with on-chip common-mode feedback (CMFB)
- Rail-to-rail input: common-mode 0.1 V to 1.7 V at $A_{OL} \geq 97$ dB
- Open-loop gain 100.3 dB (typ), GBW 7.58 MHz
- Class-AB output stage
- THD 2.8×10^{-5} % at 1 kHz, 1 V_{pp} (typ)
- Power dissipation 0.71 mW
- On-chip bias generator; only two external bias voltages required
- Verified post-layout (parasitic-extracted) across process corners and temperature

2 Description

GCOp-amp is a fully-differential operational amplifier using a folded-cascode topology with a Class-AB output stage, implemented in the SkyWater SKY130A process at a 1.8 V supply. A complementary (NMOS + PMOS) input pair provides a common-mode input range close to both rails. The output common-mode level is regulated by a CMFB loop employing source degeneration of the error pair, which remains stable across all process corners and the specified temperature range.

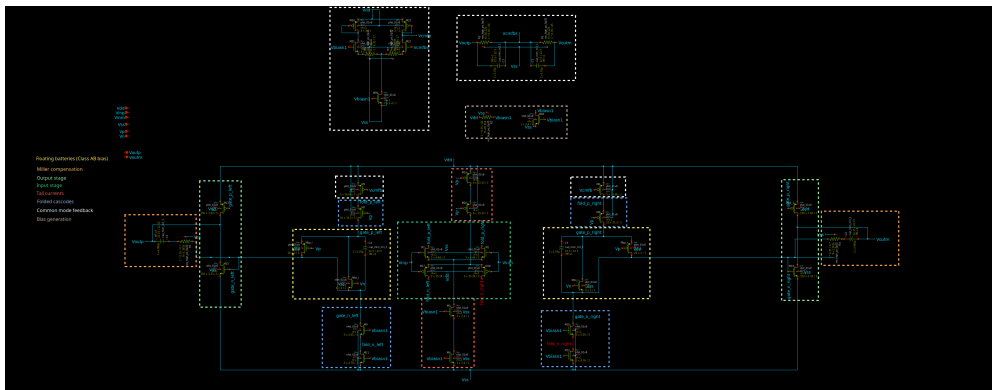
3 Functional Block Diagram



▼
 Folded cascode (XM7/XM8/XM37/XM38 PMOS,
 XM9/XM10 NMOS, XM11/XM12 sinks)
 |
 ▼
 Class-AB bias (XMbp_l/r, XMbn_l/r)
 Output banks: 10 x mult=15, PMOS + NMOS per side → Voutp
 Miller compensation XC1/XC2 + XR3/XR4 → Voutm
 | sense: XR1/XR2 + XC3/XC7
 ▼
 CMFB error pair XM21/XM22 with degeneration XR6/XR7
 actuator XM5/XM6

 Bias generator: XR5 + XM60 (diode-connected) → Vbiasn1

Figure 0. Complete Schematic



Complete Schematic

4 Pin Configuration

Pin	Type	Description
Vdd	Supply	Positive supply, 1.8 V
Vss	Supply	Ground
Vinp	Input	Non-inverting input
Vinm	Input	Inverting input
Voutp	Output	Positive output
Voutm	Output	Negative output
Vp	Input	PMOS cascode bias (0.3 V)
Vn	Input	Class-AB bias (1.5 V)

5 Recommended Operating Conditions

Parameter	Min	Typ	Max	Unit
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Parameter	Min	Typ	Max	Unit
Supply voltage	—	1.8	—	V
Operating temperature (1)	-20	—	125	°C
Input common-mode range	0.1	0.9	1.7	V
Differential load resistance	1	—	—	kΩ
Load capacitance (2)	—	1	—	pF

1. Limited by differential phase margin $\geq 45^\circ$. The CMFB loop is stable from -20 °C (Section 8).
2. Characterised at CL = 1 pF; capacitive load was not swept.

6 Electrical Characteristics

Vdd = 1.8 V, T = 27 °C, corner tt, CL = 1 pF, post-layout extracted netlist, unless otherwise noted. Min/Max are the extremes observed across process corners (tt/ss/ff/sf/fs) and temperature. All values are simulation results, not production-tested limits.

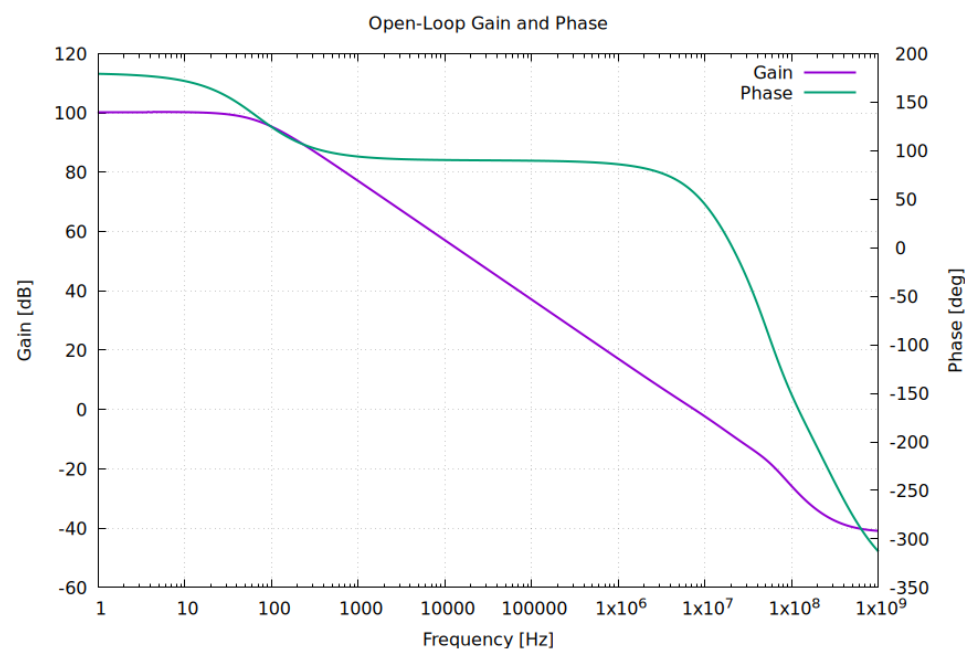
Parameter	Conditions	Min	Typ	Max	Unit
Open-loop gain A_OL	RL = ∞	89.8	100.3	108.2	dB
Open-loop gain	RL = 5 kΩ	—	79.4	—	dB
Gain-bandwidth product		4.98	7.58	9.47	MHz
Phase margin (3)		21.3	57.0	64.3	°
CMRR	DC, CM = 0.9 V	164.6	195.6	207.8	dB
PSRR+	DC	—	150.0	—	dB
PSRR–	DC	—	[N/A]	—	dB
Input offset voltage	Monte Carlo, 1 sigma	—	1.77	—	mV
Input bias current		—	0	—	A
Input noise density	1 kHz	—	264.2	—	nV/ √Hz
Input noise density	10 kHz	—	102.7	—	nV/ √Hz
Slew rate	RL = 5 kΩ	—	±3.35	—	V/μs
Overshoot	50 mV step, fs	—	0.0	—	%
Settling time (0.1 %)	50 mV step, fs	—	2.48	—	μs

Parameter	Conditions	Min	Typ	Max	Unit
THD	1 kHz, 1 Vpp	9.9e-06	2.8e-05	0.00086	%
THD	100 kHz, 1 Vpp	9.5e-05	0.0022	0.083	%
Output swing (4)	RL = 1 kΩ, single-ended	0.357	—	1.748	V
Output swing, differential (4)		—	±1.39	—	V
Output common- mode		0.913	1.049	1.159	V
Quiescent current		239	393	1101	μA
Power dissipation		—	0.707	—	mW

3. Small-signal AC result. The large-signal step response remains smooth and free of ringing in all corners (Section 8).
4. Defined as the output range for which THD remains below 1 %.

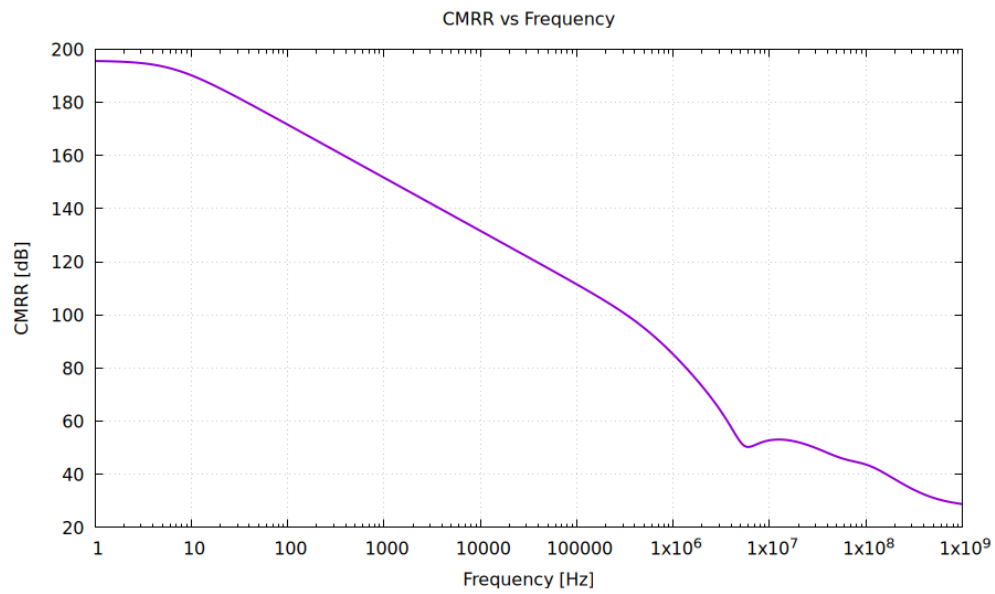
7 Typical Characteristics

Figure 1. Open-Loop Gain and Phase vs Frequency



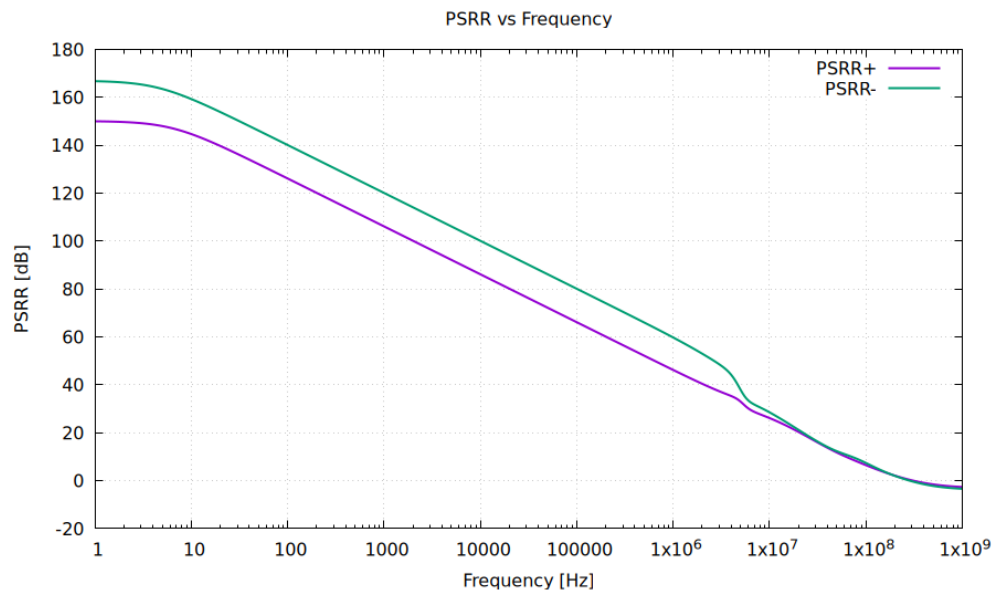
Open-Loop Gain and Phase vs Frequency

Figure 2. CMRR vs Frequency



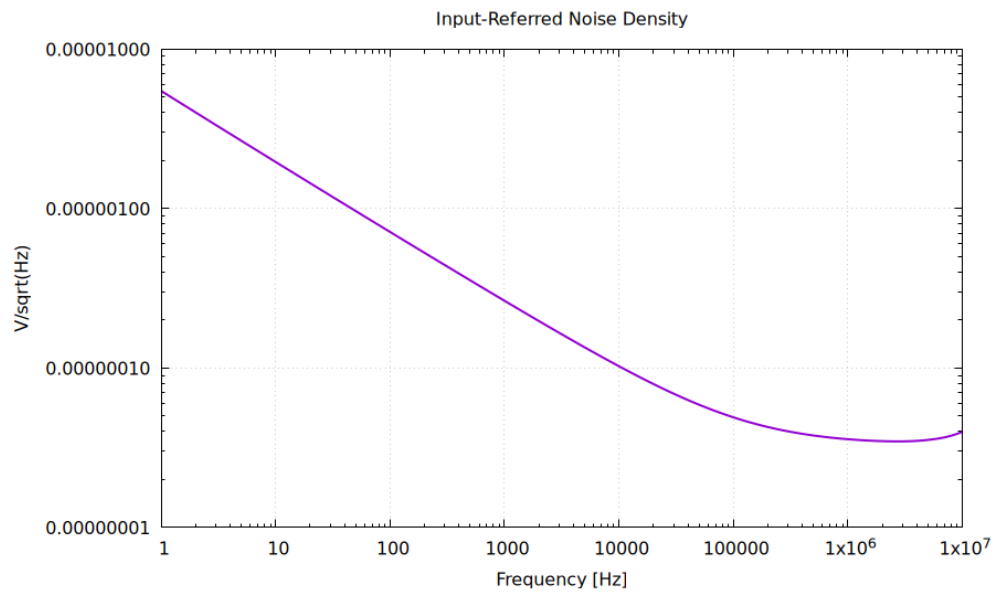
CMRR vs Frequency

Figure 3. PSRR vs Frequency



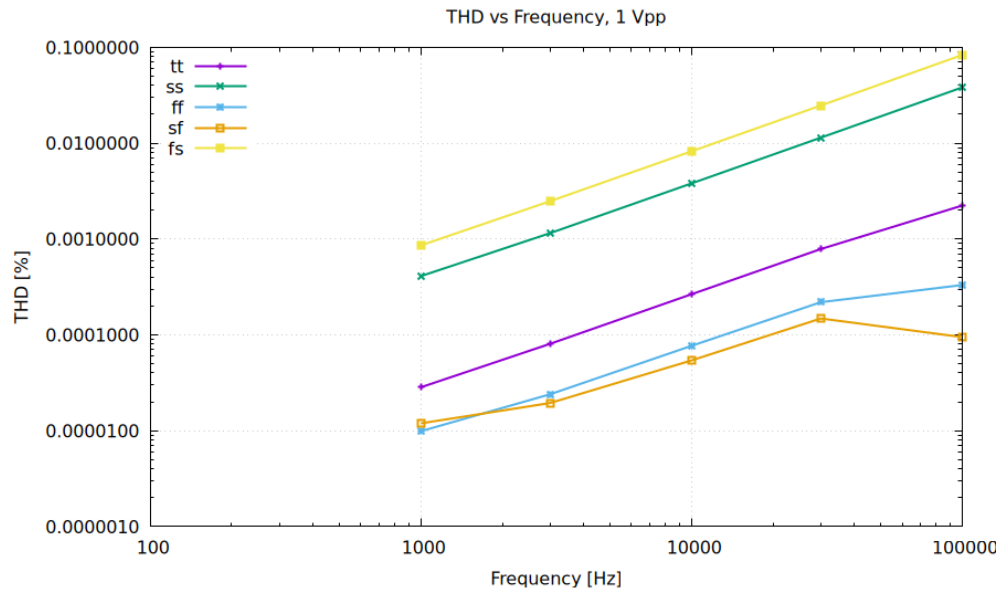
PSRR vs Frequency

Figure 4. Input-Referred Noise Density vs Frequency



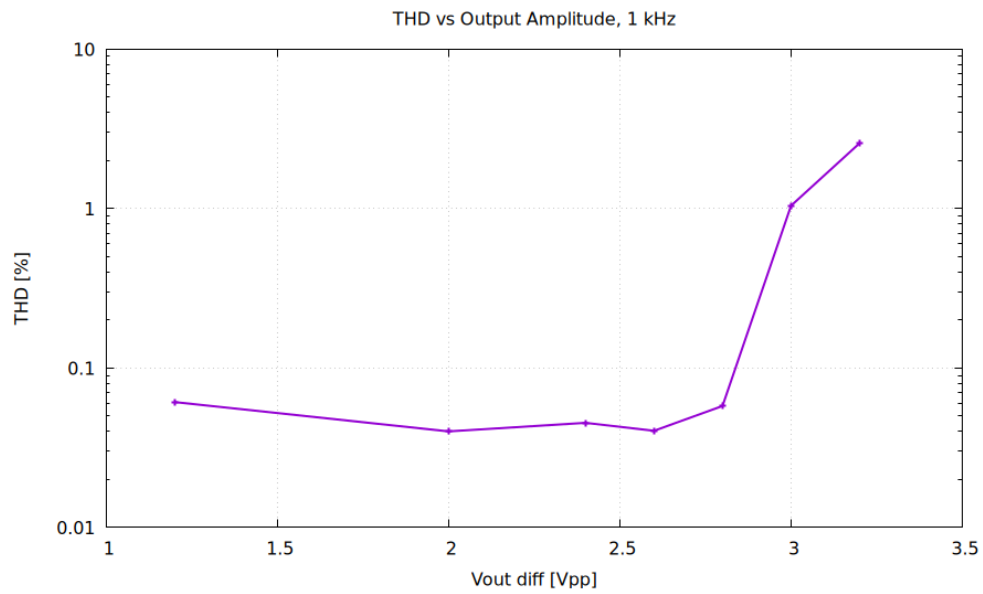
Input-Referred Noise Density vs Frequency

Figure 5. THD vs Frequency, All Corners



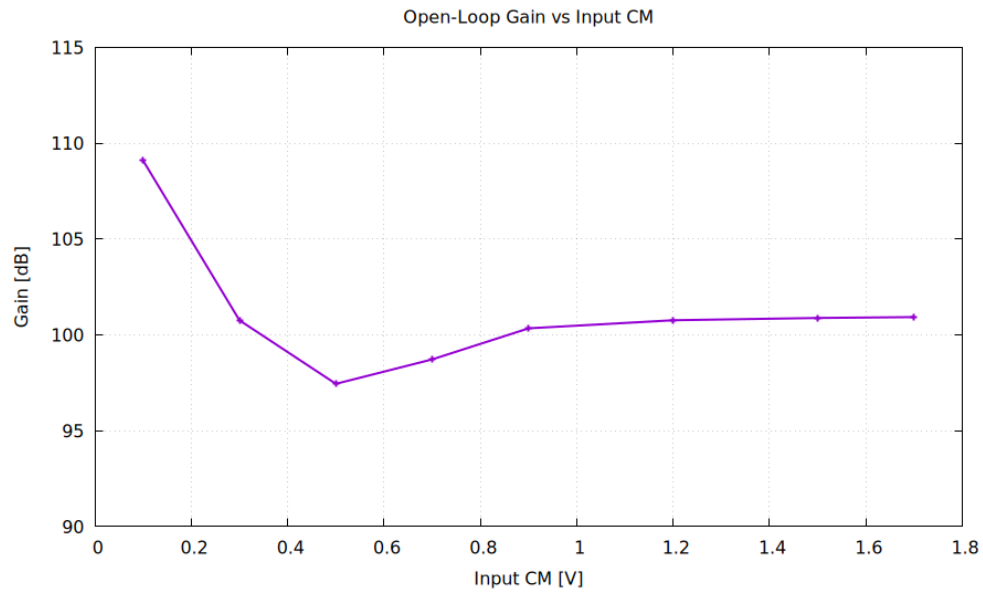
THD vs Frequency, All Corners

Figure 6. THD vs Output Amplitude at 1 kHz



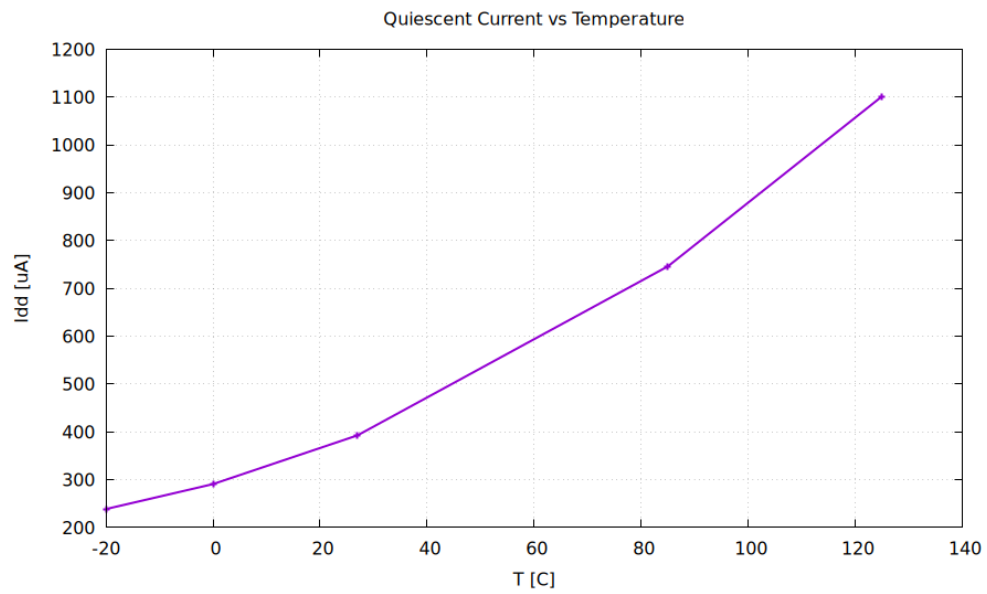
THD vs Output Amplitude at 1 kHz

Figure 7. Open-Loop Gain vs Input Common-Mode



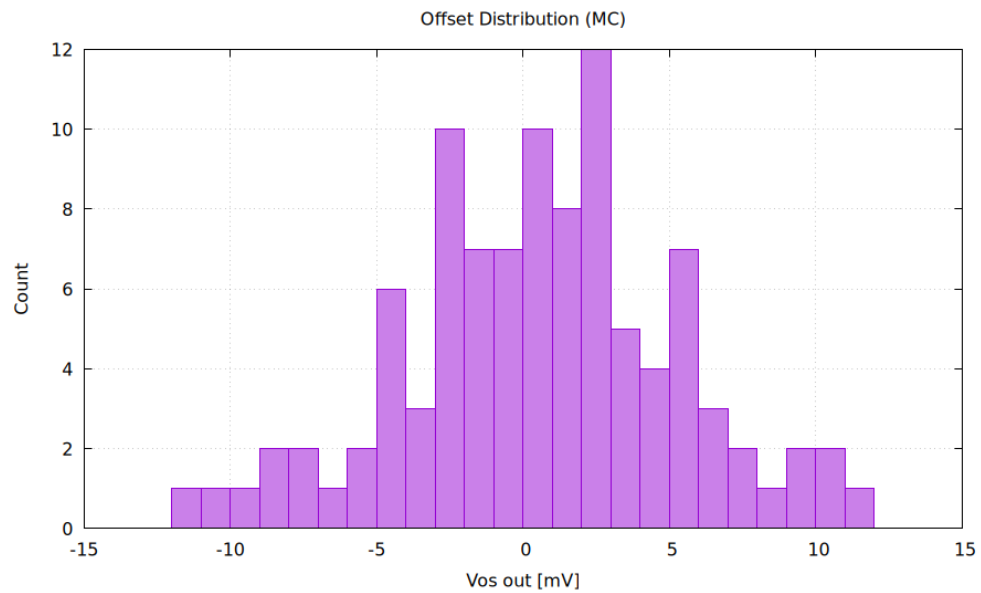
Open-Loop Gain vs Input Common-Mode

Figure 8. Quiescent Current vs Temperature



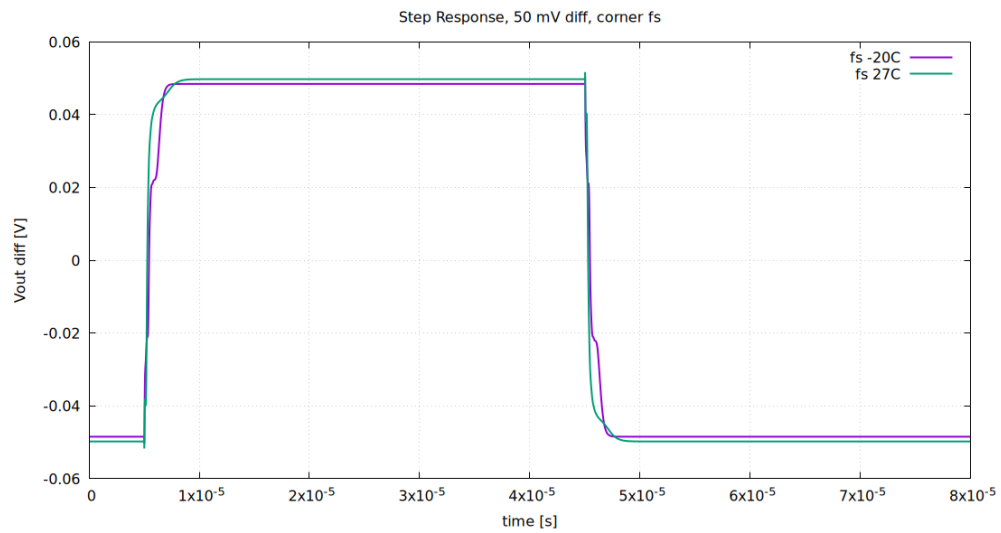
Quiescent Current vs Temperature

Figure 9. Input Offset Distribution, Monte Carlo



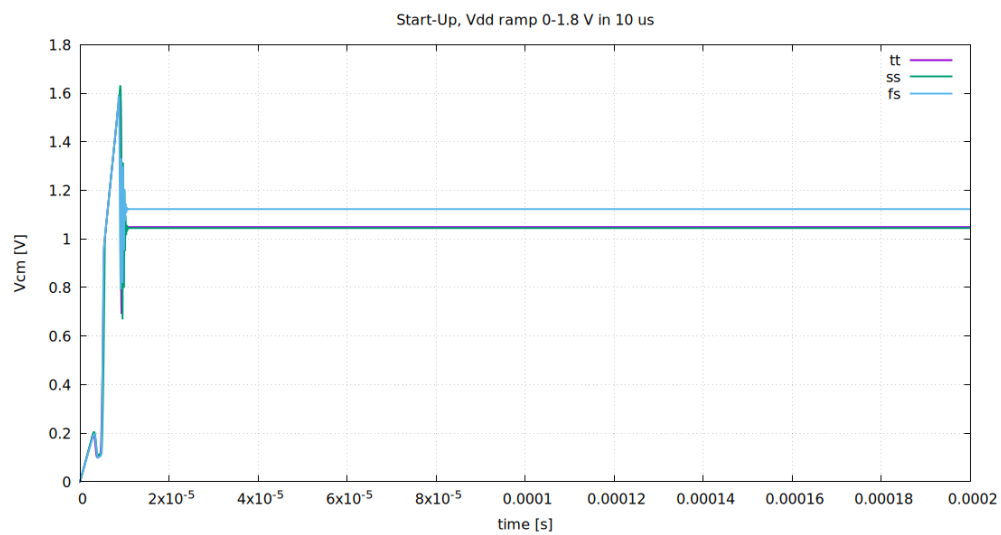
Input Offset Distribution, Monte Carlo

Figure 10. Step Response, fs Corner



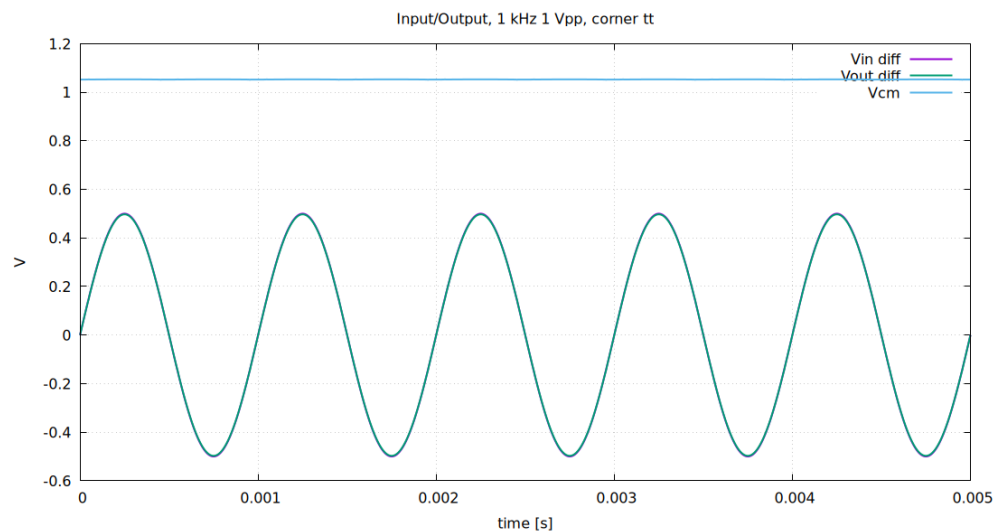
Step Response, fs Corner

Figure 11. Start-Up Behaviour, Supply Ramp



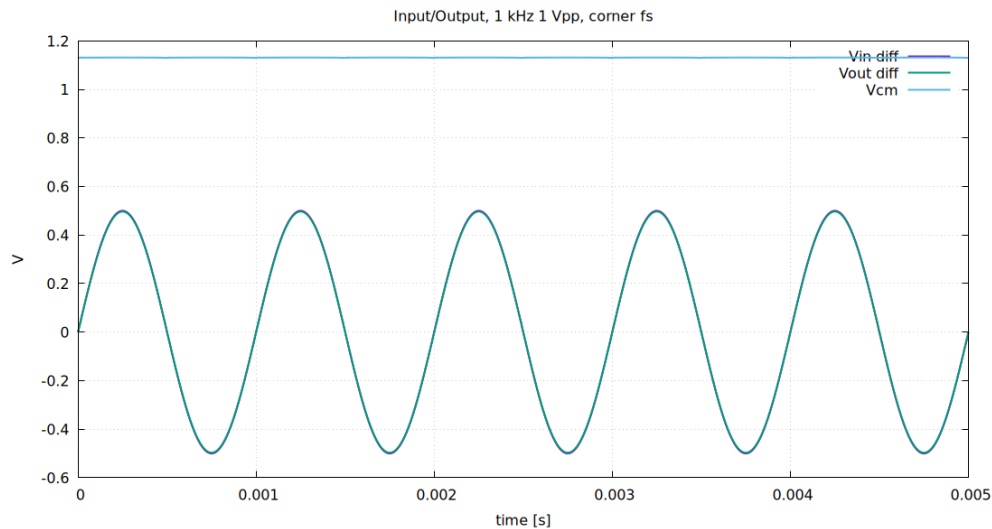
Start-Up Behaviour, Supply Ramp

Figure 12. Input/Output Waveforms, 1 kHz, Unity Gain



Input/Output Waveforms, 1 kHz, Unity Gain

Figure 13. Input/Output Waveforms, fs Corner



Input/Output Waveforms, fs Corner

8 Application Information

8.1 Closed-Loop Verification

The amplifier was verified in an inverting unity-gain configuration ($R_f = R_{in} = 10\text{ k}\Omega$, feedback factor 0.5) driving a $1\text{ k}\Omega$ differential load. The measured closed-loop gain deviates from unity by an amount consistent with the finite loop gain at that load.

8.2 CMFB Loop Stability

The CMFB loop was verified by transient analysis over $-20\text{ }^{\circ}\text{C}$ to $125\text{ }^{\circ}\text{C}$ in all five process corners: stable in all simulated corner/temperature combinations. Stability criterion: peak-to-peak deviation of the output common-mode below $1\text{ }\mu\text{V}$ after settling.

Corner	-20 °C	0 °C	27 °C	85 °C	125 °C
tt	PASS	PASS	PASS	PASS	PASS
ss	PASS	PASS	PASS	PASS	PASS
ff	PASS	PASS	PASS	PASS	PASS
sf	PASS	PASS	PASS	PASS	PASS
fs	PASS	PASS	PASS	PASS	PASS

8.3 Known Limitations

1. **Differential phase margin below 45°** outside the recommended

temperature range; worst case 21.3° (corner fs, -20 °C). This is a small-signal AC result; the transient step response remains smooth and shows no ringing even at this operating point.

2. **The output is not rail-to-rail.** The swing is bounded by the output common-mode level rather than by the output devices.
3. Common-mode input at 0.1 V is not supported in the FF and FS corners (CMFB latch).

All parameters are obtained from post-layout simulation (ngspice, R+C parasitic extraction). Transient analyses use the Gear integration method; independence from the integration timestep was verified. This document is generated automatically by `gen_datasheet.py`.